

The Moduli of the ℓ_r Basis in $\text{FBL}^{(p)}[\ell_r]$ for $r, p > 2$

Automatic math-discovery packet

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Source problem

The source paper is T. Oikhberg, M. A. Taylor, P. Tradacete, and V. G. Troitsky, *Free Banach lattices*, arXiv:2210.00614. Remark 9.48 states that for $r \geq 2$ and $1 \leq p \leq 2$, the moduli of the ℓ_r basis in $\text{FBL}^{(p)}[\ell_r]$ are equivalent to the ℓ_s basis with $1/s = 1/r + 1/2$, and then asks what happens for $r, p \in (2, \infty)$.

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Remark 9.48. Suppose $r \geq 2$ and $1 \leq p \leq 2$. Corollary 9.47 implies that the moduli of the ℓ_r basis in $\text{FBL}^{(p)}[\ell_r]$ and in $\text{FBL}[\ell_r]$ are equivalent; by [20], both are equivalent to the ℓ_s basis, with $1/s = 1/r + 1/2$. We do not know what the span of these moduli is for $r, p \in (2, \infty)$.

On the other hand, it follows immediately from Proposition 9.30 that:

Corollary 9.49. Suppose E, F are infinite dimensional Banach spaces, $p \in [1, \infty]$, $q \in (2, \infty]$, and $p \neq q$. Then $\text{FBL}^{(p)}[E]$ is not lattice isomorphic to $\text{FBL}^{(q)}[F]$.

Remark 9.50. By Corollary 9.25, if E^* has finite cotype, then $\text{FBL}[E]$

Figure 1: Remark 9.48 of the source paper, giving the known $p \leq 2$ case and asking for $r, p > 2$.

Result

Theorem 1. Let $2 < r, p < \infty$, and let (e_k) be the canonical basis of ℓ_r . Put

$$\frac{1}{s} = \frac{1}{r} + \frac{1}{p}.$$

Then the sequence $(|\delta_{e_k}|)$ in $\text{FBL}^{(p)}[\ell_r]$ is equivalent to the unit vector basis of ℓ_s . Equivalently, there are constants $c_{r,p}, C_{r,p} > 0$ such that for every finitely supported scalar sequence (a_k) ,

$$c_{r,p} \left(\sum_k |a_k|^s \right)^{1/s} \leq \left\| \sum_k a_k |\delta_{e_k}| \right\|_{\text{FBL}^{(p)}[\ell_r]} \leq C_{r,p} \left(\sum_k |a_k|^s \right)^{1/s}.$$

Idea of the proof

The free-lattice expression is nonlinear in the variables e_k , but on the dual side it is exactly the p -summing norm of a diagonal map. For $a = (a_k)_{k=1}^n$, the function

$$f_a = \sum_{k=1}^n a_k |\delta_{e_k}|$$

evaluated at $x^* = (b_k) \in \ell_{r'}$ is

$$f_a(x^*) = \sum_{k=1}^n a_k |b_k| = \|\text{diag}(a_k)b\|_{\ell_1^n}$$

when $a_k \geq 0$. Thus the defining norm of $\text{FBL}^{(p)}[\ell_r]$ is the absolutely p -summing norm of $\text{diag}(a_k) : \ell_{r'}^n \rightarrow \ell_1^n$. Garling's diagonal theorem then computes this norm as $\|a\|_{\ell_s}$, where $1/s = 1/r + 1/p$.

Proof

We first record the norm reduction. Let $a_1, \dots, a_n \geq 0$, and define

$$D_a : \ell_{r'}^n \longrightarrow \ell_1^n, \quad D_a(b_1, \dots, b_n) = (a_1 b_1, \dots, a_n b_n).$$

For $f_a = \sum_{k=1}^n a_k |\delta_{e_k}|$, the functional representation of $\text{FBL}^{(p)}[E]$, as used in Proposition 6.1 of the source paper, gives

$$\|f_a\|_{\text{FBL}^{(p)}[\ell_r]} = \sup \left\{ \left(\sum_{i=1}^m \left(\sum_{k=1}^n a_k |x_i^*(e_k)| \right)^p \right)^{1/p} : \sup_{\|x\|_{\ell_r} \leq 1} \left(\sum_{i=1}^m |x_i^*(x)|^p \right)^{1/p} \leq 1 \right\}.$$

Identifying $(\ell_r)^*$ with $\ell_{r'}$, the last display is precisely

$$\pi_p(D_a : \ell_{r'}^n \rightarrow \ell_1^n),$$

the absolutely p -summing norm of D_a .

We now apply Garling's theorem on diagonal mappings between sequence spaces: for $2 < r, p < \infty$, equivalently $1 < r' < 2 < p < \infty$,

$$\pi_p(D_a : \ell_{r'}^n \rightarrow \ell_1^n) \asymp_{r,p} \left(\sum_{k=1}^n a_k^s \right)^{1/s}, \quad \frac{1}{s} = \frac{1}{r} + \frac{1}{p}.$$

Combining the two displays proves the asserted estimate for non-negative coefficients.

Finally, the canonical basis of ℓ_r is 1-suppression unconditional. Proposition 5.2 of the source paper therefore implies that $(|\delta_{e_k}|)$ is unconditional in $\text{FBL}^{(p)}[\ell_r]$, with a constant depending only on the convention used there. Hence replacing (a_k) by $(|a_k|)$ changes the norm by at most a constant independent of n and of a . The same ℓ_s estimate follows for arbitrary scalar coefficients.

Verification and limitations

This resolves the specific $r, p \in (2, \infty)$ span question in Remark 9.48, assuming the quoted form of Garling's diagonal theorem. It is still a partial packet for the source paper as a whole: it does not address the paper's other open questions on basic sequences, POE- p , LWCG spaces, or isomorphism of free Banach lattices.

The exponent matches the endpoints suggested by the source paper. At $p = 2$ the formula gives $1/s = 1/r + 1/2$, the known case. As $p \rightarrow \infty$, it tends formally to $s = r$, matching the AM-space behavior of $\text{FBL}^{(\infty)}[\ell_r]$, where unconditional bases have equivalent moduli.

Novelty check

The run’s lightweight indexes were searched for arXiv:2210.00614, the paper title, free Banach lattice terms, and the ℓ_r -moduli language; no existing packet or attempt for this question was found. External web searches were made for the phrases

- *FBL $\hat{p}$ ell. $_r$ $p \geq 2$ moduli,*
- *we do not know what the span of these moduli is,*
- *Free Banach lattices Garling p -summing.*

The relevant hits were the source paper and the earlier Aviles–Tradacete–Villanueva paper for the non- p -convex free Banach lattice case. No later paper explicitly answering Remark 9.48 was found in this bounded search. The result should therefore be treated as a likely new identification, subject especially to verification of the exact Garling diagonal theorem statement.

Human-review recommendation

Recommended check: verify the Garling citation and exponent carefully, because the source paper notes typographical issues in Garling’s Theorem 9. Once that diagonal theorem is confirmed, the reduction to the free-lattice norm is direct.

References

- [1] T. Oikhberg, M. A. Taylor, P. Tradacete, and V. G. Troitsky, *Free Banach lattices*, arXiv:2210.00614.
- [2] D. J. H. Garling, *Diagonal mappings between sequence spaces*, Studia Math. **51** (1974), 129–138.
- [3] A. Aviles, P. Tradacete, and I. Villanueva, *The free Banach lattices generated by ℓ_p and c_0* , Rev. Mat. Iberoam. **35** (2019), 1887–1902; arXiv:1806.02553.